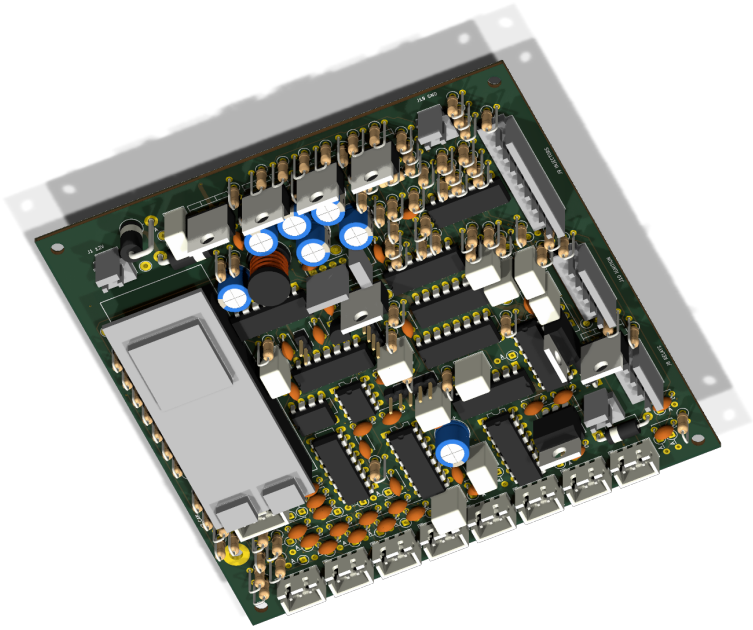
Connect. Measure. Calibrate.

Sensor installation, connection diagnosis and ECU Studio configuration

120 × 120 mm through-hole carrier • Plug-in ESP32-S3 DevKitC-1 • Revision A3-120 / 30 October 2026



Engineering prototype and bench monitor. All actuator commands remain OFF. This package does not implement a complete replacement engine ECU and has not been tested on a car. CAD checks and software tests do not establish surge immunity, EMC, splash resistance, emissions compliance or mechanical fit.

Read in this order

01 Identify the board revision, connector orientation and power arrangement.

02 Assemble and test the unpowered board, then qualify the power rails on a current-limited bench supply.

03 Use the individual sensor chapters and identical web guides. Enter the real sensor curves, then verify independent reference points.

04 Fit the insulating tray, shield bonds and splash gasket. Check the printed fit and controlled splash resistance before choosing any installation location.

What this design supports

Implemented	Outside this release
Six conditioned 0-5 V analog inputs, 12-bit SPI ADC, diagnostic conversion and filtering.	Fuel/ignition scheduling, fuel maps, tune commissioning, knock control and engine operation.
Two conditioned 5 V Hall inputs, missing-tooth crank bench decoder and cam edge counts.	Raw VR sensors, full 720° phase decoding, every OEM trigger pattern.
500 kbit/s listen-only CAN frame counts.	OEM CAN messages, dashboard, immobilizer, automatic transmission or air-conditioning integration.
External wideband-controller analog input and buffered post-catalyst measurement.	Raw wideband drive/heater control, catalyst efficiency diagnostics and OBD emissions readiness.
Design provisions for surge clamping, reverse polarity, filtering, ground plane and shield return.	Verified ISO automotive transient/EMC qualification or universal compatibility.

Universal carrier, vehicle-specific interface

01 There is no universal OEM ECU connector or sensor pinout. Use a documented adapter harness with verified cavity numbers for the specific vehicle.

02 A naturally aspirated engine with a catalytic converter can still require electronic throttle, immobilizer and network functions absent here. Keep the original ECU in service until an actual replacement system is separately engineered and qualified.

03 Do not share sensor signals with another ECU until input loading, grounding and failure behavior have been reviewed. Do not remove or emulate catalyst monitoring.

Basic bench kit

01 Current-limited regulated 12 V supply, suitable 1 A external fuse and an accessible disconnect.

02 Digital multimeter, magnifier, soldering iron and appropriate leaded soldering tools. Use a properly referenced or isolated oscilloscope for digital edges and noise.

03 3.3 V UART adapter; do not use 5 V TTL, RS-232 levels or an adapter power lead.

04 A trusted low-voltage signal/reference source and sensor-specific mechanical/reference tools. Never use a test lamp on a signal input.

Build and inspect the carrier

01 Use hardware/a3-compact, not the larger A2 board. The copper outline is 120 × 120 mm, six copper layers, 1.6 mm nominal FR-4. Check the final ERC/DRC and audit reports before ordering.

02 All carrier component joints are plated through-hole. Fit ordinary axial resistors upright on the indicated 2.54 mm pitch; R150 stays horizontal. Observe the vertical diode polarity drawing, not just the direction of a bent lead.

03 Fit low components first, then capacitors, direct-solder DIP ICs, TO-220 devices and connectors. IC sockets are not required. Leave DNP optional pull-ups absent unless the specific sensor needs them.

04 Fit the two 1 × 22 female headers for the specified 44-pin ESP32-S3 DevKitC-1 v1.1. Use a known straight header jig while soldering. Do not substitute a similar-looking 30/38-pin ESP32 or assume every S3 board has the same spacing.

05 U1 is a purchased assembled Pololu D36V6F5 module. U3 is a purchased DevKit. These modules contain small factory-soldered parts; no bare S3 module or fine-pitch IC must be hand-soldered to the carrier.

06 Check IC notches, electrolytic polarity, diode bands, transistor pin order and live TO-220 tabs against the BOM and footprints. VNP35N07-E tabs are drain, not ground.

07 Inspect every joint, bridge and solder splash. Trim solder tails to at most 3 mm below the PCB, especially around H4. The printed H4 support includes reliefs matching R35 and D210.

08 With power disconnected, check for unintended rail shorts. Capacitors can make a continuity buzzer sound briefly; investigate sustained low resistance rather than relying on the buzzer alone.

The 120 mm layout is dense. A 3D model may omit an unavailable manufacturer model; the BOM, assembly drawing and native footprint pads define the build. DNP resistors are intentionally empty. Never infer that a pad is unused simply because a 3D body is absent.

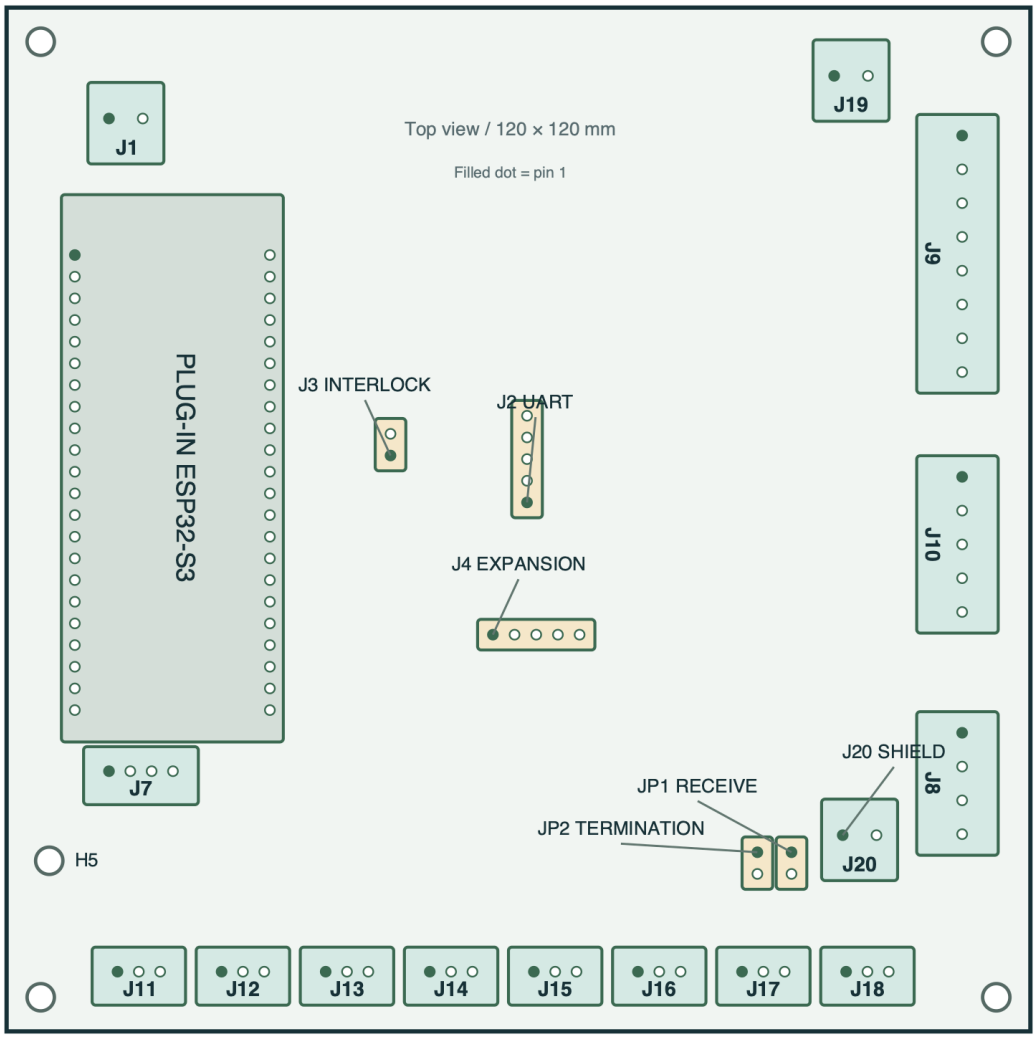
EMI provisions are an assembly requirement

01 Retain the two continuous inner ground planes and the input filter/clamps. Keep sensor returns separate from the actuator power-return wiring.

02 Use twisted/shielded signal wiring as appropriate to the sensor. Keep it away from ignition coils, alternator wiring and relay/motor power.

03 Fit the conductive enclosure lining and short internal shield bonds, then cover the lining with the supplied dielectric tray and lid insulator. A plastic box alone is not an RF shield.

Find the connector before wiring



Connector locations from the native PCB, viewed from the top. Filled dots identify pin 1. Dimensions in the table are measured from the top-left PCB edge. The separate assembly.svg shows all component references and pads.

PCB location (mm)	Connection
Bottom row, left to right	J11 MAP / J12 TPS / J13 CLT / J14 IAT / J15 wideband / J16 buffered rear oxygen / J17 crank / J18 cam
Right edge, top to bottom	J9 injector low sides / J10 smart-coil logic / J8 relay coils
Top left / top right	J1 logic power / J19 power return
Left lower side	J7 CAN and H5 shield lug
Inside board	J2 UART at 61,58 / J3 interlock at 45,52.5 / J4 expansion at 57,73.5 / J20 shield at 98,97

Check power before sensors

01 Pin 1 is externally fused key-on 12 V; pin 2 is ground. Use the specified external 1 A source fuse.

02 For first bench checks, unplug every sensor and actuator, keep J3 open, use a 12 V current-limited supply, and check polarity before enabling it.

03 Measure TP2 VIN_PROT, TP3 5 V, TP4 3V3, then each sensor pin 2. TP2 is after the reverse diode and before the input filter.

04 Repeated resets can be undervoltage/current limit, a short, wrong part orientation, regulator instability or watchdog timeout. Disconnect external loads one at a time.

Point	Meaning	Expected bench observation
TP1	GND	Reference for board checks.
TP2	VIN_PROT	Input after reverse diode, before input filter. At a 12 V source it is lower than the source by the diode drop.
TP3	+5V	Near nominal 5 V; verify under the intended sensor load.
TP4	+3V3	Near nominal 3.3 V; measure U6 VREF accurately for calibration.
TP5	SAFE_N	Near 3V3 when released; low during supervisor/watchdog reset.
TP6	RUN_5V	Low with J3 open and configuration monitor firmware.

If a rail is wrong

01 Disconnect external harnesses and remove the DevKit, then locate the fault from the input toward the load. Check part value/orientation, the reverse diode and buck module pinout.

02 If 5 V collapses only with a sensor plugged in, check its actual pin order and current draw. The sensor rail PTC is shared, slow and temperature-dependent, not a precision electronic current limiter.

03 If 3V3 resets or overheats during Wi-Fi, inspect regulator dissipation, supply limits, capacitor/ESR requirements and current peaks. Software compilation does not verify power integrity.

04 Scope rails, SAFE_N and WDI together for repeating resets. Do not bypass the watchdog or supervisor. The monitor must keep servicing supervision while the web service runs.

J1 powers the ECU logic. Injector, smart-coil and relay power is external, fused and relay-switched. Do not connect an engine or real actuator harness during these checks.

Open ECU Studio

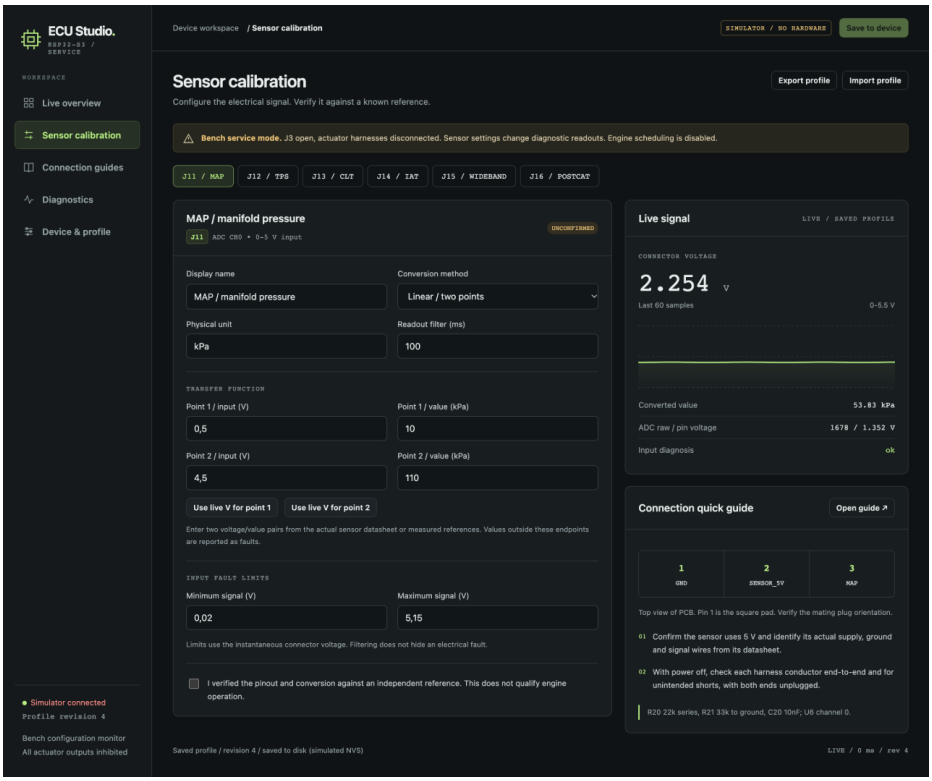
01 Flash the compact_a3_web firmware using PlatformIO. Remove the DevKit from the carrier before powering its USB connector. Disconnect USB before reseating it in the unpowered carrier.

02 Keep J3 open and actuator harnesses disconnected. Attach J2 UART using 3.3 V logic at 115200 baud, 8-N-1, with no adapter power wire.

03 Power the carrier on a dry bench with the lid open. The service starts its own WPA2 AP, ECU-Service-XXXX. A new random password is printed in the UART boot log.

04 Join that AP and open http://192.168.4.1. No internet connection is needed. The AP is refused if J3 is closed at service startup. The conductive lining may block radio reception with the lid closed.

05 For a desktop preview, run python3 tools/web_server.py and open http://127.0.0.1:8765. The simulator is labelled NO HARDWARE and uses synthetic ADC values.



Screenshot: the actual offline UI running against the host simulator. Example numbers shown are not a vehicle calibration.

Save and preserve settings

01 Export a JSON backup before editing. Import stages a complete validated profile in the editor; it does not apply until saved.

02 Save validates all channels and commits NVS before changing the live conversion. Check the saved revision and status message.

03 A stale revision is rejected instead of overwriting another session. Export edits, reload the latest profile and reapply the intended change.

04 Unreadable or invalid saved storage activates defaults and blocks saving. Inspect UART and recover deliberately; firmware does not erase NVS automatically.

Configure a measured transfer curve

01 Voltage mode shows the reconstructed connector voltage; it does not require a sensor transfer curve.

02 Linear mode uses two measured input voltages and two physical values. Set the actual units and validate a third reference point.

03 Table mode interpolates between 2-16 ascending voltage/value pairs. Values outside the table range are faults, not extrapolated measurements.

04 NTC beta mode is for CLT/IAT only and requires the physical pull-up. It uses the measured sensor supply and compensates the 55 kohm divider load. A voltage table is preferable when the real curve is available.

05 Filtering is for readout smoothing, not a substitute for fixing wiring/noise. Electrical fault checks use the instantaneous input.

06 A saved sensor profile changes diagnostic readout only. Engine scheduling, DBW, injector tuning, catalyst diagnostics and OEM protocols are not implemented.

07 Trigger geometry is applied to the bench decoder when configuration is saved. Crank synchronization is 360 degrees only.

A practical calibration sequence

01 Choose the sensor connector, verify supply/ground and compare displayed connector voltage with a trusted meter. Correct wiring before changing a curve.

02 Measure the actual ADC reference and enter it under Device & profile. A reference error changes every reconstructed voltage, including battery and sensor supply.

03 Choose voltage, two-point linear, voltage table or supported NTC beta. Enter physical units consistently. Use the real manufacturer curve or independently measured references, never the illustrative simulator curve.

04 Set an input fault window appropriate to that specific sensor. The default window is a starting point, not a diagnosis of every open/short. A legitimate endpoint can be zero volts on some interfaces.

05 Choose only as much readout filtering as necessary. Inspect raw voltage and the oscilloscope signal while checking wiring noise.

06 Save, verify several independent points across the intended range and only then mark the calibration confirmed. Record the sensor part number, curve source, pull-up and reference tools.

Mode	Fields and behavior
Voltage	Reconstructs connector volts: ADC / 4095 × measured VREF / 0.6.
Linear	V0/Y0 and V1/Y1; ascending voltage endpoints. Physical values can increase or decrease. Outside the endpoints is a fault.
Table	2-16 ascending voltage/value pairs. Piecewise linear interpolation. No extrapolation beyond the table.
NTC beta	CLT/IAT only. R0, reference temperature, beta and actual physical pull-up. Uses measured sensor supply and 55 kohm load compensation.
Fault/filter	Fault checks use instantaneous voltage. The display filter smooths readout, not engine control.

Sensor connector pin reference

J11-J18 use the same PCB-side three-pin assignment. This does not establish the sensor-side wire colors or pin order. The mating view may be mirrored.

Connector	Signal / channel	Pin 1	Pin 2	Pin 3
J11	MAP / manifold pressure / ADC 0	GND	SENSOR_5V	MAP
J12	TPS / throttle position / ADC 1	GND	SENSOR_5V	TPS
J13	Coolant temperature / ADC 2	GND	SENSOR_5V	CLT
J14	Intake-air temperature / ADC 3	GND	SENSOR_5V	IAT
J15	Wideband controller output / ADC 4	GND	SENSOR_5V	LAMBDA_WB
J16	Buffered post-catalyst signal / ADC 5	GND	SENSOR_5V	POSTCAT_BUF
J17	Crank Hall / GPIO8	GND	SENSOR_5V	CRANK_RAW
J18	Cam Hall / GPIO9	GND	SENSOR_5V	CAM_RAW

Before connecting any sensor

01 Board pin numbers refer to the PCB top view; pin 1 is the square pad. The mating plug view can be mirrored. Verify the actual harness with continuity tests.

02 Disconnect power before resistance/continuity tests. Do not measure resistance on a live vehicle bus or use a megohmmeter on electronics.

03 Keep J3 open and all actuator harnesses disconnected during sensor commissioning. This firmware keeps ARM off.

04 Use the actual sensor or controller manufacturer pinout, transfer curve, mounting torque and air gap. There is no universal vehicle-side pinout.

05 Measure signal voltage against that sensor connector pin 1, not an arbitrary chassis bolt. Never use a test lamp on a sensor or MCU pin.

Passive two-wire CLT/IAT NTCs use pins 1 and 3 only. Wideband-controller analog outputs use pins 1 and 3; pin 2 does not power a heater/controller. Raw VR and raw LSU sensors are unsupported without the appropriate external interface.

MAP / manifold pressure

PIN 1	PIN 2	PIN 3
1 GND	2 SENSOR_5V	3 MAP

Mounting & routing

01 Retain the OEM pressure port or use a suitable sealed manifold-pressure takeoff after the throttle. Keep the hose short and secure, and follow the sensor maker's fluid/orientation limits.

02 Route away from hot exhaust and ignition wiring. Leak-check any added hose before running an engine.

Connect & verify

01 Confirm the sensor uses 5 V and identify its actual supply, ground and signal wires from its datasheet.

02 With power off, check each harness conductor end-to-end and for unintended shorts, with both ends unplugged.

03 Power the board, check pin 2 near 5 V, then connect the sensor and measure pin 3 versus pin 1.

04 Use a regulated hand vacuum/pressure source within the sensor rating, compare the web voltage with a meter, then enter the manufacturer transfer curve or measured reference points.

Expected behavior

01 Key-on / engine-off pressure should agree with a trusted local absolute-pressure reference after calibration. It is not always 100 kPa.

02 A steady applied pressure should produce a steady voltage and calibrated value.

Fault tracing

01 Flat near 0 V: inspect power, ground, signal short and sensor pin order.

02 Flat high: inspect a short to supply, wrong curve, or a saturated sensor.

03 Pressure wrong but voltage agrees with the meter: check absolute versus gauge units, curve and pressure reference.

04 Noisy or slow: inspect hose leaks/restriction, ground offset, shield routing and configured filtering.

R20 22k series, R21 33k to ground, C20 10nF; U6 channel 0.

TPS / throttle position

PIN 1	PIN 2	PIN 3
1 GND	2 SENSOR_5V	3 TPS

Mounting & routing

01 Use the OEM TPS mounting and its correct shaft coupling. A single analog input does not replace an electronic throttle system or its redundant safety channels.

02 Turn the throttle only with the vehicle made safe; do not force an electronic throttle plate or alter mechanical stops.

Connect & verify

01 Identify whether the TPS is a 3-wire potentiometer or another sensor. Do not connect throttle-motor wires to J12.

02 Verify pin 2 supply and pin 1 ground, then observe signal voltage while moving the permitted mechanism slowly.

03 Record closed and fully open reference voltages, enter those as 0% and 100%, and mark calibration confirmed only after checking the whole travel.

Expected behavior

01 The voltage should sweep smoothly without dropouts. Some sensors have a decreasing transfer curve; configure the corresponding endpoint values.

Fault tracing

01 Jumps at one position: inspect sensor track/coupling or a broken harness conductor.

02 Direction reversed: swap calibration endpoint values, not supply and ground wires.

03 Does not reach 0%/100%: recheck mechanical travel and measured endpoints.

04 Two TPS tracks or a throttle motor are present: this hardware does not implement OEM DBW control.

R22 22k series, R23 33k to ground, C21 10nF; U6 channel 1.

Coolant temperature

PIN 1	PIN 2	PIN 3
1 GND	2 SENSOR_5V	3 CLT

Mounting & routing

01 Use the OEM coolant sensor and mounting boss, with the engine cold and cooling system depressurized. Follow the specified seal and torque.

02 Keep the original vehicle sensor if required by other systems; verify loading before sharing any signal.

Connect & verify

01 For a passive two-wire NTC, use pins 1 and 3 only. Leave pin 2 empty in the mating housing.

02 Fit R32 only if this is a passive NTC requiring the board pull-up. Check its actual resistance; the design value is 2.49 kohm.

03 Measure sensor resistance while disconnected and compare at known temperatures with its datasheet. Prefer several reference points.

04 Enter beta parameters or a voltage table. In NTC mode, confirm the physical pull-up is fitted; the UI cannot install a resistor.

Expected behavior

01 A common NTC resistance falls as temperature rises, and its signal voltage falls with this pull-up topology.

02 The conversion accounts for the board's 55 kohm divider load and measured sensor supply.

Fault tracing

01 Near 0 V: sensor/loom short to ground or very low resistance.

02 High signal / open-model fault: unplugged sensor, broken ground/signal or missing/wrong pull-up.

03 Plausible voltage but wrong temperature: wrong NTC curve, pull-up value, supply reading or units.

04 Do not assume a floating/unexcited input will read maximum voltage: R32 is DNP by default.

R24 22k series, R25 33k to ground, C22 10nF; U6 channel 2.

R32 is DNP by default. Fit it only after confirming the actual sensor topology and resistor value.

Intake-air temperature

PIN 1	PIN 2	PIN 3
1 GND	2 SENSOR_5V	3 IAT

Mounting & routing

01 Use a suitable intake-air boss and the sensor maker's orientation. Avoid placing the sensing element against the intake wall or at an unrepresentative heat source.

02 Follow the original mechanical arrangement; secure any aftermarket sensor so it cannot enter the intake.

Connect & verify

01 A passive two-wire NTC connects to pins 1 and 3; pin 2 remains empty.

02 R33 is the optional board pull-up, normally 2.49 kohm. Confirm it is physically fitted before using NTC conversion.

03 Compare disconnected resistance and stabilized readings with a reference thermometer at several temperatures. Enter the actual sensor curve.

Expected behavior

01 After a long cold soak, intake and coolant temperature should be reasonably consistent with their actual surroundings. Heat soak can produce real differences.

Fault tracing

01 Use the same open/short/pull-up checks as CLT, but inspect intake heat soak and airflow before blaming calibration.

02 Fast spikes suggest intermittent wires or electrical interference. A slow heat-soak drift may be physical.

R26 22k series, R27 33k to ground, C23 10nF; U6 channel 3.

R33 is DNP by default. Fit it only after confirming the actual sensor topology and resistor value.

Wideband controller output

PIN 1	PIN 2	PIN 3
1 GND	2 SENSOR_5V	3 LAMBDA_WB

Mounting & routing

01 Install the exhaust sensor according to the sensor/controller maker's rules for bung angle, temperature and condensation. The ECU board does not heat or drive a raw wideband sensor.

02 Mount the external controller away from hot exhaust and splash paths; use its specified fused power and grounds.

Connect & verify

01 Connect controller analog signal to pin 3 and its specified analog reference to pin 1.

02 Leave pin 2 unpopulated; the board sensor supply must not power the controller or heater.

03 Enter the controller's actual 0-5 V mapping and units. Lambda and AFR are not interchangeable without the correct fuel stoichiometry.

04 Use the controller's documented analog-output test mode or a known voltage source before interpreting exhaust readings.

Expected behavior

01 Web voltage and a meter should agree; converted lambda should agree with the controller when both use the same mapping and ground reference.

Fault tracing

01 Controller display correct but ECU wrong: inspect mapping, analog-reference offset and output mode.

02 Fixed fault voltage: read the external controller's error codes before treating it as mixture.

03 No output until heated: verify controller state and heater supply.

04 A raw LSU sensor cannot be connected directly; do not invent a direct 5-wire/6-wire pin mapping.

R28 22k series, R29 33k to ground, C24 10nF; U6 channel 4.

Buffered post-catalyst signal

PIN 1	PIN 2	PIN 3
1 GND	2 SENSOR_5V	3 POSTCAT_BUF

Mounting & routing

01 Retain the catalyst and rear oxygen sensor in their proper locations. This input is for measurement, not catalyst emulation or bypass.

02 Use a suitable external high-input-impedance conditioner/buffer where needed. The board input presents about 55 kohm to the sensor signal.

Connect & verify

01 Connect only the buffer's conditioned 0-5 V output to pin 3 and its analog reference to pin 1. Leave pin 2 empty unless the buffer specifically permits this supply within the shared current budget.

02 Do not connect heater wires, a raw wideband sensor, or assume every raw narrowband output can tolerate this load.

03 Compare the buffer output with a meter; enter its transfer curve or retain voltage display.

Expected behavior

01 Voltage is the directly supported diagnostic. This firmware does not implement catalyst-efficiency monitoring, emissions readiness or OEM rear-sensor diagnostics.

Fault tracing

01 Signal changes when shared with OEM ECU: stop and review input loading and the buffer interface.

02 Zero or clipped output: inspect buffer supply/reference and output limits.

03 A stable rear reading alone does not prove catalyst health.

R30 22k series, R31 33k to ground, C25 10nF; U6 channel 5.

Crank Hall input

PIN 1	PIN 2	PIN 3
1 GND	2 SENSOR_5V	3 CRANK_RAW

Mounting & routing

01 Use the original rigid bracket/target-wheel relationship and the exact manufacturer air gap, polarity, supply rating and torque. There is no universal trigger gap.

02 This input accepts a compatible 5 V Hall interface. A two-wire variable-reluctance sensor needs an external conditioner.

Connect & verify

01 Verify supply and ground before connecting. Determine whether the Hall output is push-pull or open-collector.

02 For an open-collector output without another suitable pull-up, fit the specified optional pull-up and verify the current/voltage requirements.

03 Use an oscilloscope to compare connector pin 3 with the conditioned MCU input. A multimeter can miss missing pulses and noise.

04 With ignition/injection disabled, turn the target safely or use an isolated bench signal generator. Do not connect a generator in parallel with an energized sensor.

Expected behavior

01 Only rising edges are captured. Both signal paths use two Schmitt inversions, preserving the input polarity.

02 Crank sync requires the selected missing-tooth geometry and repeated valid gaps. Cam is counted but full 720-degree phase decoding is not implemented.

Fault tracing

01 No edges: inspect supply, missing pull-up, wrong sensor type, target gap, signal/ground pin order and U7 conditioning.

02 Edges but no sync: inspect total slots/missing teeth, speed range, signal quality and electrical noise. A simple square wave is not a valid missing-tooth wheel.

03 False RPM / overflow: inspect ringing, shield/return routing, gap and pull-up; power-cycle after a latched capture overflow.

R40 33k series, C30 100pF, optional R41 2.2k pull-up; U7 -> GPIO8.

Cam Hall input

PIN 1	PIN 2	PIN 3
1 GND	2 SENSOR_5V	3 CAM_RAW

Mounting & routing

01 Use the original rigid bracket/target-wheel relationship and the exact manufacturer air gap, polarity, supply rating and torque. There is no universal trigger gap.

02 This input accepts a compatible 5 V Hall interface. A two-wire variable-reluctance sensor needs an external conditioner.

Connect & verify

01 Verify supply and ground before connecting. Determine whether the Hall output is push-pull or open-collector.

02 For an open-collector output without another suitable pull-up, fit the specified optional pull-up and verify the current/voltage requirements.

03 Use an oscilloscope to compare connector pin 3 with the conditioned MCU input. A multimeter can miss missing pulses and noise.

04 With ignition/injection disabled, turn the target safely or use an isolated bench signal generator. Do not connect a generator in parallel with an energized sensor.

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02 Edges but no sync: inspect total slots/missing teeth, speed range, signal quality and electrical noise. A simple square wave is not a valid missing-tooth wheel.

03 False RPM / overflow: inspect ringing, shield/return routing, gap and pull-up; power-cycle after a latched capture overflow.

R42 33k series, C31 100pF, optional R43 2.2k pull-up; U7 -> GPIO9.

Power, data and output connections

J1 / Logic power

01 Pin 1 is externally fused key-on 12 V; pin 2 is ground. Use the specified external 1 A source fuse.

02 For first bench checks, unplug every sensor and actuator, keep J3 open, use a 12 V current-limited supply, and check polarity before enabling it.

03 Measure TP2 VIN_PROT, TP3 5 V, TP4 3V3, then each sensor pin 2. TP2 is after the reverse diode and before the input filter.

04 Repeated resets can be undervoltage/current limit, a short, wrong part orientation, regulator instability or watchdog timeout. Disconnect external loads one at a time.

J2 / UART / boot / reset

01 Pins: 1 ground, 2 ECU TX -> adapter RX, 3 ECU RX <- adapter TX, 4 EN, 5 BOOT_N. Use 3.3 V logic, 115200 baud, 8-N-1.

02 Do not connect an adapter power wire or a powered DevKit USB cable while the carrier supplies the module. For USB programming, remove the module from the carrier.

03 If silent, check supply and EN, cross TX/RX correctly, verify the A3 build and chosen serial port, then capture the boot log. The existing console tag can still say ecu_a0.

04 For ROM download mode, hold BOOT_N low during an EN reset, then release it. Do not short EN or BOOT_N to 12 V.

Power, data and output connections

J3 / Physical interlock

01 Pin 1 is 3V3 and pin 2 is INTERLOCK. Open is the normal state during commissioning.

02 lock=0 with the contact open is expected. Closing it can allow hardware run gating; do not close it with vehicle actuators connected.

03 The configuration firmware keeps ARM=0 regardless. Never bypass SAFE_N to make a fault disappear.

J4 / Internal expansion

01 Pins: ground / 3V3 / SDA / SCL / SAFE_N. This is an internal 3.3 V service bus, not a 12 V sensor connector.

02 Check pull-ups and idle SDA/SCL near 3V3. A stuck-low line suggests a short, wrong device voltage or a device holding the bus.

03 No generic external I2C scanner or expansion actuator control is implemented by this release.

Power, data and output connections

J7 / JP1 / JP2 / CAN and termination

01 J7: CANH / CANL / ground / CHASSIS. Keep the pair twisted and verify the network bitrate. Firmware reception is 500 kbit/s listen-only; no OEM decoding is provided.

02 JP1 open leaves MCP2562 in standby, so ordinary frames will not be received. Close it only on the appropriate supervised bus for normal transceiver reception. Firmware remains listen-only.

03 JP2 adds onboard 120 ohm termination when closed. Leave it open on an already terminated vehicle network.

04 With the complete network unpowered, two 120 ohm end terminations normally measure about 60 ohm between CANH and CANL. A removed/unpowered interface alone need not read 60 ohm.

05 No frames: inspect JP1, bitrate, polarity, grounds, bus activity and termination. A standalone two-node listen-only bench cannot supply missing ACKs for a lone transmitter.

J8 / External relay coils

01 Pins: MAIN low side / PUMP low side / FAN low side / ground. External fused relay supplies and correctly oriented flyback diodes are required.

02 Target coil current is at most 150 mA per channel, pending physical tests. Do not connect a fan motor or pump motor directly.

03 The configuration firmware commands all channels OFF. An inactive relay is expected, not a reason to bypass the inhibit.

04 With a coil connected to a safe test supply, OFF low-side voltage can be near the external supply, not zero. Driver/tab is electrically live. Actuation tests require a separate qualified dummy-load procedure.

Power, data and output connections

J9 / J19 / Injector outputs and power return

01 J9 pins 1-4 are injector low sides 1-4; pins 5-8 are ground. J19 both pins are the dedicated power return.

02 Only saturated high-impedance PFI injectors are in scope; 1.2 A/channel is a design target pending tests. No GDI or peak-and-hold driver is provided.

03 Actuator 12 V is external, fused and relay switched. Never use the sensor ground harness as the injector current return.

04 All outputs remain OFF in this firmware. Check resistance/continuity only disconnected; do not pulse a fueled engine to troubleshoot wiring.

05 A high OFF output with an externally powered load is normal. Persistent current with ARM off indicates wrong wiring, a shorted driver or a board fault: disconnect power.

J10 / Smart-coil logic only

01 Pins 1-4 are active-high 5 V smart-coil commands; pin 5 is ground. They are not ignition-coil primary outputs.

02 Use only a compatible low-current logic input (design target at most 4 mA). Coil power and power-ground wiring remain external.

03 The configuration firmware keeps dwell commands OFF. Do not use a test lamp, an ordinary LED without its resistor, or a coil primary as a logic test load.

04 Changing or interrupting dwell on a real coil can produce a spark. Keep real ignition hardware disconnected during bench configuration.

Power, data and output connections

J20 / H5 / Shield and enclosure bond

01 J20 pins and plated lug H5 are CHASSIS, not normal sensor/actuator current returns. Bond them to the internal lining using the assembly guide.

02 Check lining/braid continuity with power off. The CHASSIS-to-GND 1 Mohm bleed and 4.7 nF RF capacitor are intentional, not a hard DC ground link.

03 Keep all bonds inside the water seal and route any external reference conductor through a sealed gland. Do not drill an unsealed shield-stud hole.

TP1-TP6 / Test points

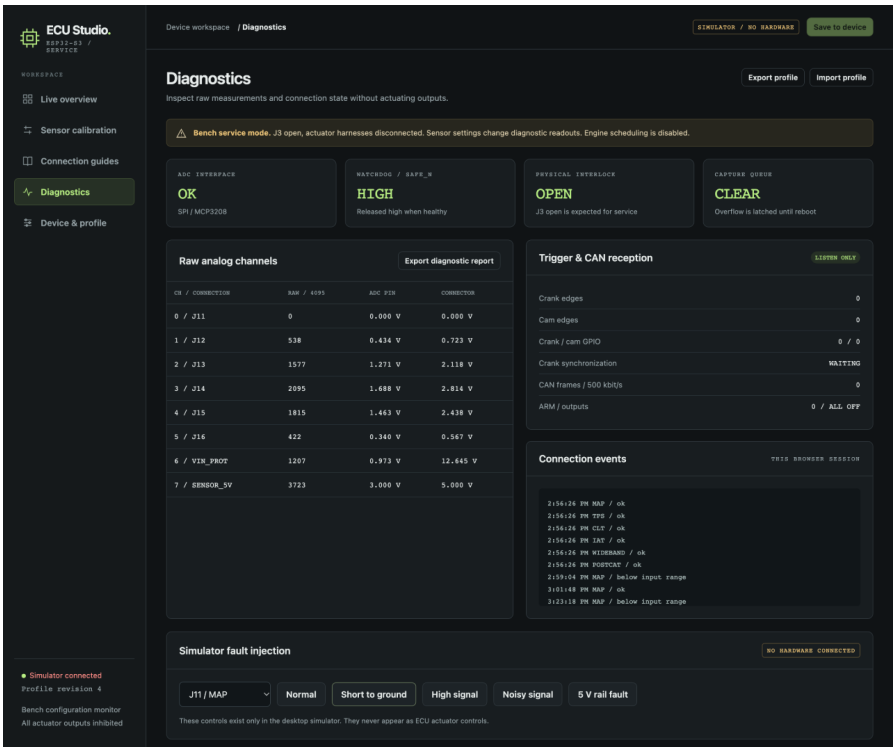
01 TP1 ground; TP2 VIN_PROT; TP3 5 V; TP4 3V3; TP5 SAFE_N; TP6 RUN_5V.

02 With J3 open and monitor firmware, RUN_5V should be low even when supplies are healthy. SAFE_N is an open-drain supervised net, normally pulled near 3V3 when released.

03 If SAFE_N repeatedly drops, scope it with the supply rails and WDI. Do not defeat the MAX1232 or 3V3 supervisor.

04 Use a properly referenced or isolated scope arrangement; do not clip an earth-referenced scope ground to an arbitrary live output.

Trace a fault from the connector inward



Screenshot: host simulator with a deliberately shorted MAP input. Fault-injection controls are absent from the embedded firmware.

01 Start with battery and sensor-rail measurements. If several channels fail together, inspect shared power/ground before adjusting every curve.

02 Compare pin 3 versus pin 1 at the connector with the web connector voltage. If those agree, inspect curve, units and reference settings.

03 If connector voltage is correct but ADC is wrong, check the named 22k/33k input divider, clamp polarity, filter capacitor, U6 reference and SPI connections. Do not inject an automotive battery voltage as a signal test.

04 If all ADC channels are stuck at zero/full scale, inspect U6 power/reference, chip select, MOSI/MISO/clock, solder bridges and the live ADC transfer status. A successful SPI transaction alone cannot prove sensor integrity.

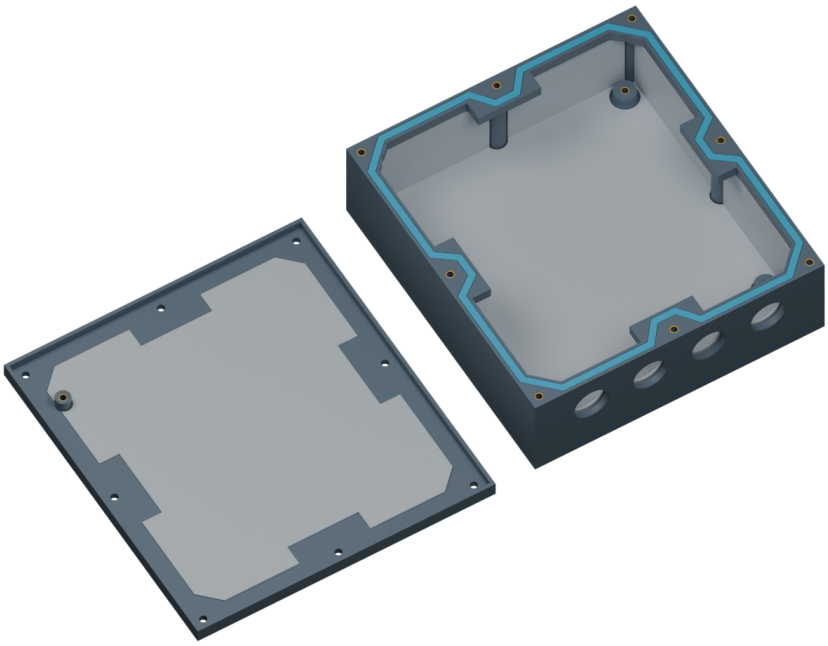
05 For missing crank/cam counts, scope both raw and conditioned signals. Review pull-ups, Hall-versus-VR type, target geometry and air gap before blaming software.

06 Capture overflow is latched until power cycle. Find the signal/queue cause first; clearing a latch does not repair the source.

07 Export the diagnostic report with timestamp, profile revision and browser event history. Event history starts when that browser session starts. The export is not a persistent ECU flight recorder.

08 If the page disconnects, readings become stale. Check the AP, board supply and UART, then reconnect. Never use a frozen browser number as a live measurement.

Install the board and shield



PCB: 120 × 120 mm. Printed base: 138 × 150 × 44 mm. Lid: 142 × 154 mm. Nominal assembled height: 47 mm. Cable glands add to the rear envelope.

01 Print base, lid, insulating tray, lid insulator and insert coupon. Use a material/process suitable for the measured installation temperature, chemicals and vibration. Do not assume a printed case is suitable beside an exhaust or underhood heat source.

02 Test the coupon with the actual short M3 heat-set inserts before fitting them to the case. Four PCB supports take M3 short inserts; eight lid bosses and two internal shield bonds use the same insert type.

03 Install the internal base shield-bond insert and lining before the PCB blocks tool access. Keep the side-wall bond blind: do not drill through the outer wall.

04 Apply conductive lining following the templates. Keep it below and clear of the water-seal land. Bond seams with a suitable conductive method and use short broad internal bonds. The gasket is not an electrical bond.

05 Fit the insulating tray over the lining. Clear the support roots and internal bond boss. Ensure conductive foil cannot touch solder tails, module pins, live transistor tabs or terminals.

06 Install the PCB at box origin 9,9 mm and Z=9 mm. Hole centers are (13,13), (125,13), (13,125), (125,125) mm from the base origin. Use four M3 × 6 screws and 0.8 mm nylon washers.

07 Trim all solder tails to at most 3 mm below the PCB. H4 has specific relief pockets for R35 and D210; align the board correctly and confirm the tail clearances by dry fitting.

08 Dry fit all mating plugs, wire bends and strain relief with the unpowered board. Do not force a cable against the lid, a live transistor tab or the gasket. The CAD component envelope does not qualify the finished harness.

09 Fit the removable lid lining bond and its dielectric cover. Leave a flexible internal bond loop that cannot snag, touch electronics or cross the gasket.

Seal the lid and cable entries

01 Cut one continuous ring from 2.0 mm soft closed-cell silicone sheet with the supplied 1:1 template. The ring is 3 mm wide; the groove is 3.4 mm wide and 1.5 mm deep. Do not print the reference gasket in ordinary rigid filament.

02 Inspect the sealing land, groove, case walls and layers for pores, strings and damage. Establish a compatible sealing/post-processing method if the printing process leaves porous walls.

03 Seat the ring without stretches, twists, gaps or adhesive lumps. The nominal geometry compresses a 2 mm sheet to 1.5 mm (25%); check the actual chosen material and print dimensions.

04 Fit four suitable M16 cable glands, or gasketed blanking plugs for unused holes. Their panel flange must fit the 24 mm diameter envelope. Use the manufacturer sealing washers and cable-jacket diameter range.

05 A single-hole gland seals against a round cable jacket, not a bundle of loose individual wires. Use appropriately jacketed multi-core harness groups and sealed external branches. The JST connectors are inside the box and are not sealed external bulkhead connectors.

06 Provide harness strain relief and a drip loop. Place cable entries away from direct splash and pooling. The rear entry centers are X=27,55,83,111 mm and Z=27 mm.

07 Close the lid with its drip skirt outside the base wall. Use eight M3 × 8 screws with 0.8 mm washers, tighten evenly and inspect continuous gasket seating. Screw holes lie outside the nominal water-seal line.

08 Test the empty assembled case with absorbent paper before installing electronics. Start with controlled light splashes in the intended orientations, inspect every seam and gland, dry completely, and correct leaks before repeating. No IP rating or pressure-wash/submersion capability is claimed.

Splash resistance, printed fit, insert strength, vibration, thermal behavior and shielding performance require physical tests. The CAD checks prove nominal solid/mesh integrity and specified clearances only. Keep all service work dry; opening the lid removes the splash barrier.

Hardware list

01 4 × M3 × 6 PCB screws and 0.8 mm nylon washers; 8 × M3 × 8 lid screws and 0.8 mm flat washers.

02 14 × ruthex RX-M3Sx4.0 short inserts: 4 PCB, 8 lid, 2 internal shield bonds. Check the actual insert datasheet and printed coupon.

03 2 × M3 × 5 bond screws with suitable ring terminals/star washers; H5 uses a separate M3 bolt/nut and short braid.

04 2 mm closed-cell silicone gasket sheet, conductive lining/bond materials, four matched M16 glands or sealed blanks.

Record the evidence for this build

Complete this record for the actual board, harness and sensors. Blank fields mean not checked. A browser screenshot does not replace a measurement.

Item	Record / result
Board serial, PCB revision and assembly date	_____
Firmware environment / hash and profile revision	_____
12 V source, fuse and current-limit setting	_____
TP2 / TP3 / TP4 and measured U6 VREF	_____
Each sensor part number, actual pinout and curve source	_____
CLT/IAT pull-up values and independent temperature points	_____
MAP pressure reference / TPS travel reference points	_____
External wideband controller and analog-output mapping	_____
Crank/cam waveform, wheel pattern and air gap source	_____
CAN network state / termination / bitrate	_____
Printed material, gland/cable sizes and gasket material	_____
Dry fit, tail clearance, strain relief and bond continuity	_____
Empty-box splash test observations / orientation	_____
Outstanding faults and qualification work	_____

Separate qualification still needed

01 Measure reverse-polarity, supply transients/load dump, cranking brownout, regulator stability and thermal margins with suitable controlled equipment.

02 Qualify sensor fault stress, clamp currents, output current/clamp/thermal behavior and hardware inhibition on appropriate dummy loads.

03 Check EMC/immunity, grounding, harness interaction, vibration and environmental performance.

04 An actual vehicle ECU replacement needs a reviewed vehicle harness, complete engine control implementation and calibration, safety analysis and applicable emissions/legal assessment.

Files and manufacturer references

The same structured guide content appears in the browser and in this manual. Keep the PDF, firmware, hardware and profile schema from the same release.

`hardware/a3-compact/esp32-ecu.kicad_pcb and .kicad_sch: native design`

`hardware/a3-compact/bom.csv and connector-pinout.csv: assembly and pin contract`

`enclosure/a3/: STEP, STL, native FreeCAD, gasket/lining templates and dimensions`

`config/interface-guide.json: source of sensor mounting, checks and fault tracing`

`config/defaults.json: uncalibrated profile; schema version 1`

`docs/web-configuration.md: service setup, persistence and recovery`

`reports/web/: UI screenshots and automated software test results`

`reports/a3/: electrical CAD checks and dimensional audit`

Guide source SHA-256: e7a8ad098e3a92081bf15585b5727021d456e7c5d4296063b2cbd840c57205ad

Primary references

Espressif S3 DevKitC v1.1 pins and power methods

https://docs.espressif.com/projects/esp-dev-kits/en/latest/esp32s3/esp32-s3-devkitc-1/user_guide_v1.1.html

Microchip MCP3208 ADC

https://www.microchip.com/en-us/product/mcp3208

Microchip MCP2562 CAN transceiver

https://ww1.microchip.com/downloads/en/DeviceDoc/25167A.pdf

Bosch LSU 4.9 requires a dedicated controller

https://www.bosch-motorsport.com/products/sensors/lambda/lambda-sensor-mini-lsu-4-9/

Kvaser CAN termination checks

https://kvaser.com/developer-blog/how-to-test-your-can-termination-works-correctly/

ST protected low-side driver

https://www.st.com/resource/en/datasheet/vnp35n07-e.pdf

Always obtain the exact sensor/connector/module datasheets and the vehicle wiring diagram. These references explain component interfaces; they do not establish this prototype as vehicle-qualified.